Ventricular Fibrillation / Pulseless VT

AC-09 | Adult Cardiac

Quick flow

01 VF / Pulseless VT [EMT / AEMT / Paramedic]

Start CPR • bag-mask ventilation with oxygen • attach monitor/defibrillator • activate arrest team

02 Shock 1 [EMT / AEMT / Paramedic]

200 J biphasic • verify manufacturer setting • immediately resume CPR for 2 min

03 CPR 2 min + IV/IO [EMT / AEMT / Paramedic]

Rate 100-120/min • depth ≥2 in • full recoil • minimize pauses • IV preferred, IO if delayed

04 Shock 2 [AEMT / Paramedic]

Same or higher energy per device • immediately resume CPR • epinephrine 1 mg IV/IO ASAP, then every 3-5 min

05 CPR 2 min + Airway [EMT / AEMT / Paramedic]

Consider advanced airway/capnography without interrupting compressions • treat reversible causes

06 Shock 3 [EMT / AEMT / Paramedic]

Same or higher energy per device • immediately resume CPR

07 CPR 2 min + Antiarrhythmic [Paramedic]

Amiodarone 300 mg IV/IO; second dose 150 mg • alternative lidocaine 1-1.5 mg/kg IV/IO, then repeat per dose table (maximum total 3 mg/kg)

08 Rhythm Every 2 min [EMT / AEMT / Paramedic]

Shockable → shock and resume CPR • organized rhythm → pulse check • ROSC → AC-10

09 Persistent Arrest [EMT / AEMT / Paramedic]

Continue shock/CPR cycles • epinephrine every 3-5 min • treat Hs and Ts • consider AC-12 when criteria met

Overview

- Adult pathway for patients 16 years of age and older in ventricular fibrillation or pulseless ventricular tachycardia.
- High-quality CPR, immediate defibrillation, minimal peri-shock pauses, and treatment of reversible causes take priority over advanced procedures.
- Claiborne EMS uses an initial biphasic defibrillation energy of 200 J; confirm the manufacturer-recommended setting and use the same or higher energy for subsequent shocks according to the device.

EMT Care

Immediate CPR, AED operation, ventilation, compressor rotation, timing, and documentation.

EMT

- Confirm unresponsiveness, abnormal or absent breathing, and absence of a pulse for no more than 10 seconds; begin CPR immediately.
- Perform compressions at 100-120/min to a depth of at least 2 inches with complete recoil and minimal interruption.
- Apply the AED/monitor pads and deliver a shock immediately when advised; resume CPR without a post-shock pulse check.

- Provide bag-mask ventilation with oxygen at a 30:2 compression-to-ventilation ratio until an advanced airway is placed.
- Rotate compressors every 2 minutes or sooner when fatigued and announce cycle, shock, medication, and rhythm-check times.
- Check a pulse only when an organized rhythm appears and prepare for immediate re-arrest if ROSC occurs.

AEMT Care

All EMT care plus vascular access, epinephrine, airway assistance, and medication preparation.

AEMT

- Perform all EMT actions and establish IV access during CPR without delaying a shock; use IO access when IV access is unsuccessful or would delay medication.
- After the second shock, give epinephrine 1 mg IV/IO as soon as possible and repeat every 3-5 minutes.
- Assist with advanced-airway placement and continuous waveform capnography without interrupting compressions or delaying defibrillation.
- Prepare amiodarone, lidocaine, and magnesium sulfate for paramedic administration.
- Continuously reassess CPR quality, ventilation rate, waveform capnography, medication timing, and reversible causes.
- Do not independently administer amiodarone, lidocaine, or magnesium sulfate under AC-09.

Paramedic Care

All AEMT care plus manual defibrillation, rhythm interpretation, antiarrhythmics, advanced airway/capnography, and reversible-cause treatment.

Paramedic

- Perform all EMT and AEMT actions and confirm VF or pulseless VT while maintaining continuous high-quality CPR.
- Defibrillate at 200 J biphasic, confirm the manufacturer-recommended setting, and use the same or higher energy for subsequent shocks according to the device.
- Charge the defibrillator during compressions, clear briefly, deliver the shock, and resume CPR immediately.
- After the second shock, give epinephrine 1 mg IV/IO as soon as possible and repeat every 3-5 minutes.
- After the third shock, give amiodarone 300 mg IV/IO; give 150 mg IV/IO as the second dose for persistent or recurrent VF/pVT.
- As an alternative to amiodarone, give lidocaine 1-1.5 mg/kg IV/IO initially; then 0.5-0.75 mg/kg every 5-10 minutes to a maximum total of 3 mg/kg. Do not combine amiodarone and lidocaine without Medical Control.
- For suspected pulseless torsades, give magnesium sulfate 1-2 g IV/IO push; may repeat once. Do not use magnesium routinely for other VF/pVT arrests.
- Place an advanced airway and use continuous waveform capnography when this can be accomplished without interrupting compressions or delaying shocks.

Indications

- Ventricular fibrillation without a pulse in a patient 16 years of age or older.
- Pulseless ventricular tachycardia in a patient 16 years of age or older.
- Pulseless polymorphic ventricular tachycardia managed as a shockable cardiac-arrest rhythm.

Contraindications

- There is no contraindication to immediate defibrillation for confirmed VF or pulseless VT.
- Do not defibrillate an organized rhythm unless the patient is pulseless and the rhythm is confirmed as ventricular tachycardia.

- Do not interrupt CPR or delay defibrillation for vascular access, medication administration, or advanced-airway placement.

Assessment | Immediate arrest assessment

- Confirm unresponsiveness, abnormal or absent breathing, and absence of a pulse for no more than 10 seconds.
- Begin CPR, provide bag-mask ventilation with oxygen, attach pads, and identify VF or pulseless VT.
- Establish the time last known responsive, whether the arrest was witnessed, bystander CPR/AED use, initial rhythm, downtime, and suspected cause when this does not interrupt treatment.

Assessment | CPR quality

- Compression rate 100-120/min, depth at least 2 inches, complete recoil, and minimal interruptions.
- Change compressors every 2 minutes or sooner when fatigued.
- Keep rhythm analysis, pulse checks, and peri-shock pauses under 10 seconds.
- Charge during compressions and resume compressions immediately after every shock.

Assessment | Rhythm, ventilation, and perfusion monitoring

- Analyze rhythm every 2 minutes; check a pulse only when an organized rhythm appears.
- Before an advanced airway, use a 30:2 compression-to-ventilation ratio.
- After an advanced airway, continue compressions and give one breath every 6 seconds.
- Use continuous waveform capnography after advanced-airway placement to confirm position and monitor ventilation and CPR trends.
- An abrupt sustained rise in $EtCO_2$ may suggest ROSC but requires rhythm and pulse confirmation; no absolute $EtCO_2$ value determines termination.

Assessment | Reversible causes

- Evaluate continuously for hypovolemia, hypoxia, hydrogen ion excess/acidosis, hypo-/hyperkalemia, and hypothermia.
- Evaluate for tension pneumothorax, cardiac tamponade, toxins, pulmonary thrombosis, and coronary thrombosis.
- Use the linked renal/hyperkalemia and toxic-ingestion protocols for cause-specific treatment.

Treatment steps

- Begin high-quality CPR, provide bag-mask ventilation with oxygen, attach the monitor/defibrillator, and activate the arrest team.
- Deliver the first shock at 200 J biphasic after confirming the manufacturer-recommended setting; immediately resume CPR for 2 minutes.
- During the first post-shock CPR cycle, establish IV access; use IO access if IV access is unsuccessful or would delay medication.
- Perform a rhythm check after 2 minutes. If VF/pVT persists, deliver the second shock using the same or higher energy according to the device and immediately resume CPR.
- After the second shock, give epinephrine 1 mg IV/IO as soon as possible and repeat every 3-5 minutes.
- Consider advanced-airway placement and continuous waveform capnography without interrupting compressions or delaying defibrillation.
- After the next 2-minute CPR cycle, deliver the third shock if VF/pVT persists and immediately resume CPR.
- After the third shock, give amiodarone 300 mg IV/IO. If VF/pVT persists or recurs, give a second dose of 150 mg IV/IO.

- Lidocaine 1-1.5 mg/kg IV/IO may be used as the initial alternative to amiodarone; then give 0.5-0.75 mg/kg every 5-10 minutes to a maximum total of 3 mg/kg. Do not combine amiodarone and lidocaine without Medical Control.
- For suspected pulseless torsades, give magnesium sulfate 1-2 g IV/IO push; may repeat once. Do not give magnesium routinely for other VF/pVT arrests.
- Continue 2-minute CPR cycles, rhythm checks, shocks for persistent VF/pVT, epinephrine every 3-5 minutes, and treatment of reversible causes.
- If an organized rhythm appears, briefly check for a pulse. ROSC requires immediate transition to AC-10; persistent arrest may transition to AC-12 when criteria are met.

Medications

Epinephrine

1 mg IV/IO after the second shock as soon as possible; repeat every 3-5 minutes

- AEMT and Paramedic standing order under AC-09; no EMT administration.
- Do not interrupt CPR or delay defibrillation for medication administration.
- Document each dose and maintain a visible 3-5-minute medication timer.

Amiodarone

First dose 300 mg IV/IO; second dose 150 mg IV/IO

- Paramedic standing order for VF/pVT refractory to the third shock.
- Administer during CPR without interrupting compressions.
- Do not combine with lidocaine without Medical Control.

Lidocaine

1-1.5 mg/kg IV/IO initial dose; then 0.5-0.75 mg/kg every 5-10 minutes; maximum total 3 mg/kg

- Paramedic alternative to amiodarone for shock-refractory VF/pVT.
- Administer during CPR without interrupting compressions.
- Do not combine with amiodarone without Medical Control.

Magnesium sulfate

1-2 g IV/IO push; may repeat once

- Paramedic standing order only for suspected pulseless torsades/long-QT polymorphic VT.
- Routine magnesium is not indicated for other VF/pVT cardiac arrests.

Warnings / precautions

- Do not delay defibrillation for airway placement, vascular access, medication, scene movement, or a 12-lead ECG.
- Resume CPR immediately after every shock; do not perform a routine post-shock pulse check.
- Avoid excessive ventilation. After advanced-airway placement, give one breath every 6 seconds with continuous compressions.
- Do not combine amiodarone and lidocaine without explicit Medical Control direction.
- Routine magnesium, calcium, and sodium bicarbonate are not indicated in undifferentiated VF/pVT; use them only for a specific reversible cause or linked protocol indication.
- No absolute $EtCO_2$ value should be used alone to terminate resuscitation.
- Routine vector-change defibrillation and double-sequential defibrillation are not included because their usefulness remains unestablished.

Clinical pearls

- Defibrillation and high-quality CPR are the highest-priority treatments for VF/pVT.

- Charging during compressions and immediate post-shock CPR reduce harmful pauses.
- A rhythm check occurs every 2 minutes, but a pulse check occurs only when an organized rhythm appears.
- An abrupt sustained $EtCO_2$ increase may indicate ROSC; confirm with an organized rhythm and pulse without an unnecessary interruption.
- VF/pVT may be the result of coronary occlusion, hyperkalemia, hypothermia, toxins, or another reversible cause that requires simultaneous treatment.

Special populations | Suspected pulseless torsades

- Defibrillate and continue standard VF/pVT arrest care.
- Give magnesium sulfate 1-2 g IV/IO push; may repeat once.
- Open AC-08 if a pulse returns with recurrent polymorphic VT.

Special populations | Suspected hyperkalemic or toxicologic arrest

- Continue standard VF/pVT care while opening the linked cause-specific protocol.
- Calcium and sodium bicarbonate are reserved for specific indications rather than routine cardiac-arrest administration.

Related pathways / contacts

AC-03 Adult Cardiac Arrest - Open the master adult cardiac-arrest pathway.

AC-08 Polymorphic Wide-Complex Tachycardia - Open if a pulse returns with recurrent polymorphic VT.

AC-10 Post-Resuscitation Care - Open immediately after return of spontaneous circulation.

AC-11 TEAM-Focused CPR - Open for team roles, timing, communication, and CPR quality.

AC-12 Termination of Resuscitation - Open for persistent arrest when termination criteria may be met.

AM-03 Dialysis / Renal Failure - Open for suspected hyperkalemic or dialysis-related arrest.

TE-07 Overdose / Toxic Ingestion - Open for suspected sodium-channel blocker, opioid, or other toxicologic arrest.

Source and references

American Heart Association. 2025 Adult Cardiac Arrest Algorithm. | American Heart Association. 2025 Guidelines for CPR and ECC: Adult Advanced Life Support. | Tennessee Emergency Medical Services Board. Tennessee EMS Protocol Guidelines, September 2025. | Claiborne EMS medication formulary and related cardiac, renal, and toxicology protocols.

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